Gaurav Mahajan

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RESEARCH INTERESTS — Reinforcement Learning; Post-Training LLMs; Learning Theory

CURRENT POSITION

Postdoctoral Researcher, Institute for Foundations of Data Science
Yale University, New Haven, CT
Advisor: Daniel Spielman

EDUCATION

Doctor of Philosophy, Computer Science
University of California, San Diego, 2023
Advisors: Sanjoy Dasgupta, Shachar Lovett

Bachelor of Science, Mathematics and Computing
Indian Institute of Technology, Delhi, 2013

WORK EXPERIENCE

University of Alberta — Summer 2022
Research Intern
Mentor: Csaba Szepesvari

Microsoft Research, New York — Summer 2021
Research Intern
Mentors: Sham Kakade, Akshay Krishnamurthy

Institute for Advanced Study, Princeton — Fall 2019
Short-Term Scholar
Mentor: Jason Lee

Simons Institute for Theory of Computing, Berkeley — Summer 2019
Visiting Graduate Student
Mentor: Jason Lee

Microsoft, Redmond — 2014 - 2017
Software Engineer

Epic Systems, Madison — 2013 - 2014
Software Engineer

Ecole Normale Superieure, Cachan — Summer 2012
Visiting Undergraduate Student
Mentor: Stefan Haar

PUBLICATIONS

Nikhil Bansal, Matthias Caro and Gaurav Mahajan. *Cloning is as Hard as Learning for Stabilizer States.* Under Submission. 2025

Adam Bene Watts, John Bostanci, Matthias Caro, Daniel Grier, Sihan Liu and Gaurav Mahajan. *Generalized de Finetti Theorem: Lifting pure state algorithms to mixed states* . Under Submission. 2025

Daniel Grier, Sihan Liu and Gaurav Mahajan. *Improved classical shadows from local symmetries in the Schur basis.* Under Submission. 2025

Max Hopkins, Daniel Kane, Shachar Lovett and Gaurav Mahajan. *Do PAC-Learners Learn the Marginal Distribution?.* ALT 2025.

Sham Kakade, Akshay Krishnamurthy, Gaurav Mahajan and Cyril Zhang. *Learning Hidden Markov Models Using Conditional Samples.* COLT 2023.

Daniel Kane, Sihan Liu, Shachar Lovett, Gaurav Mahajan, Csaba Szepesvari and Gillert Weisz. *Exponential Hardness of Reinforcement Learning with Linear Function Approximation.* COLT 2023.

Daniel Kane, Sihan Liu, Shachar Lovett and Gaurav Mahajan. *Computational-Statistical Gaps in Reinforcement Learning.* COLT 2022.

Max Hopkins, Daniel Kane, Shachar Lovett and Gaurav Mahajan. *Realizable Learning is All You Need.* COLT 2022.

Robi Bhattacharjee and Gaurav Mahajan. *Learning What To Remember.* ALT 2022.

Sanjoy Dasgupta, Gaurav Mahajan and Geelon So. *Convergence of online k-means.* AISTATS 2022.

Simon Du, Sham Kakade, Jason Lee, Shachar Lovett, Gaurav Mahajan, Wen Sun and Ruosong Wang. *Bilinear Classes: A Structural Framework for Provable Generalization in RL.* ICML 2021.

Alekh Agarwal, Sham Kakade, Jason Lee and Gaurav Mahajan. *On the Theory of Policy Gradient Methods: Optimality, Approximation, and Distribution Shift.* JMLR 2021.

Max Hopkins, Daniel Kane, Shachar Lovett and Gaurav Mahajan. *Point Location and Active Learning: Learning Halfspaces Almost Optimally.* FOCS 2020.

Alekh Agarwal, Sham Kakade, Jason Lee and Gaurav Mahajan. *Optimality and Approximation with Policy Gradient Methods in Markov Decision Processes.* COLT 2020.

Max Hopkins, Daniel Kane, Shachar Lovett and Gaurav Mahajan. *Noise-tolerant, Reliable Active Classification with Comparison Queries.* COLT 2020.

Simon Du, Jason Lee, Gaurav Mahajan and Ruosong Wang . *Q-learning with Function Approximation in Deterministic Systems.* NeurIPS 2020.

SELECTED INVITED TALKS

Learning language models using conditional queries.

- BLISS Seminar, UC Berkeley
- Foundations of Data Science Seminar, Yale University
- CSE Theory Seminar, UC San Diego

- Computer Science Seminar Series, University of Victoria
- Computer Science Department Speaker Series, Oberlin College
- Computer Science Seminar, Barnard College
- Science for AI Seminars, University of Birmingham

Computational-statistical gaps in reinforcement learning.
- Research at TTIC Seminar Series, Toyota Technological Institute at Chicago
- Big Data & Machine Learning Seminar, UCLA
- BLISS Seminar, UC Berkeley
- Foundations of Data Science Seminar, Yale University
- EnCORE Fall Retreat, UC San Diego

Equivalence between realizable and agnostic learning.
- Theory Seminar, Cornell University
- Machine Learning Reading Group, Microsoft Research

Towards a theory of generalization in reinforcement learning.
- RL Theory Seminars (virtual seminar series)
- CSE Theory Seminar, UC San Diego

TEACHING EXPERIENCE

Instructor for *Bootcamp on Reinforcement Learning Theory* (International Centre for Theoretical Sciences) — Fall 2025

Instructor for *Quantum Codes and Applications to Complexity* (Yale University) — Fall 2024

PROFESSIONAL ACTIVITIES

Mentor, Learning Theory Alliance (LeT-All) Fall 2023 Mentorship Workshop

Invited reviewer
- *Discrepancy Theory: Computational and Geometric Perspective*, a graduate-level survey to appear in *Foundations and Trends in Theoretical Computer Science* (2025)
- Conferences and journals: COLT, ALT, JMLR, ICML, ICLR, WORLIT, NeurIPS